

PakTravel AI System

Project Report

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Part 1: Route Planning (Search Algorithms)

Scenario: PakTravel operates a bus network across 15 major cities in Pakistan. Ali, a student from Karachi, is travelling to Multan for a wedding. The AI system must determine the optimal path for his journey, including potential relay routes where a bus change is required, ensuring minimum travel distance.

Methodology: The road network was modelled as an undirected weighted graph (adjacency dictionary). Three search algorithms were implemented to navigate this graph:

1. **Uniform Cost Search (UCS):** Explores paths based strictly on the lowest cumulative distance.
2. **A* Search:** Enhances UCS by incorporating a heuristic (straight-line distance to Islamabad) to guide the search directionally.
3. **Bidirectional Search:** Runs two simultaneous breadth-first searches from the start and goal to find an intersection. Additionally, a custom relay-planner was built to stitch together two separate UCS searches, and the network was visualized using networkx.

Results: All algorithms successfully routed Ali from Karachi to Multan (via Hyderabad and Sukkur). When compared on a journey to Islamabad, UCS and A* found the exact same shortest path distance. The relay planner successfully generated 2-leg routes (e.g., Karachi to Multan via Sukkur) and accurately compared their distances against direct routes.

Analysis: While UCS and A* returned identical route distances, A* proved to be more computationally efficient. By utilizing the heuristic, A* explored fewer nodes because it avoided expanding paths that moved away from the target destination (such as roads leading toward Quetta or Gwadar). Bidirectional search was fast but lacked the weight-awareness necessary for finding true shortest geographical paths.

Conclusion: A* Search is the most optimal and efficient algorithm for PakTravel's route planning. It perfectly balances finding the mathematically shortest route with minimizing computational overhead, which is critical for real-time GPS navigation.

Part 2: AI Legal Advisor (Propositional Logic)

Scenario: During the journey, a PakTravel bus driver named Ahmed is caught speeding and using a mobile phone. He has previous unpaid challans, three total violations, and is a repeat offender. The system must automatically deduce his legal penalties using formal logic.

Methodology: An expert system was built using Propositional Logic via the aima-python library. 15 real-world traffic laws were encoded into a Knowledge Base (KB) using logical implications (e.g., `Speed_Above_Limit ==> Challan_Issued`). Ahmed's specific facts were told to the KB. The system then utilized **Resolution Refutation**, converting the rules into Conjunctive Normal Form (CNF) to formally prove whether his license should be suspended or if he should be arrested.

Results: The AI Legal Advisor correctly deduced that Ahmed's license must be suspended and calculated a total fine of 2,000 PKR (for mobile phone usage). However, it ruled that Ahmed should *not* be arrested.

Analysis: The Resolution Refutation algorithm proved the license suspension through two distinct logical paths: one triggered by his "Three Violations" and another triggered by the combination of his "Speeding" (resulting in a challan) and "Not Paid" status. The system spared him from arrest because Rule 5 requires the driver to be Still_Driving after a suspension, a fact that was not present in Ahmed's case.

Conclusion: Propositional logic provides a highly deterministic, bias-free mechanism for automating legal decisions. By using Resolution, PakTravel can guarantee mathematically sound penalization for its drivers based strictly on established rules and proven facts.

Part 3: Bus Scheduling (Constraint Satisfaction Problem)

Scenario: Ali has arrived in Sukkur and needs his connecting bus to Multan (Route 3). PakTravel must assign 10 available buses to 8 specific routes without any scheduling overlaps, prioritizing large buses for highway routes and ensuring drivers get adequate rest.

Methodology: The scheduling task was modelled as a Constraint Satisfaction Problem (CSP):

- **Variables:** 8 Routes (R1 to R8)
- **Domains:** 10 Buses (B1 to B10)
- **Hard Constraints:** No simultaneous time overlaps for the same bus.
- **Soft Constraints:** +2 points for assigning large buses to highways; +1 point if a bus runs routes. The problem was solved using Plain Backtracking, enhanced with the Minimum Remaining Values (MRV) heuristic, and pre-processed with the AC-3 arc consistency algorithm.

Results: The CSP algorithms successfully generated a complete, conflict-free schedule. The optimization function achieved a perfect score (16/16) by perfectly distributing the large buses and ensuring no bus was overworked. Ali's bus (R3) was successfully assigned to a vehicle at 12pm.

Analysis: Because the domain (10 buses) was larger than the variables (8 routes) and the constraints were simple inequalities, plain backtracking found the solution almost immediately, and AC-3 did not significantly reduce the initial domains. Most importantly, the nature of the CSP guaranteed Ali's connection: a CSP is only "solved" when *all* variables (including R3) are assigned a valid value without violating the time constraints.

Conclusion: CSP modelling is exceptionally well-suited for logistics and resource allocation. By defining clear hard and soft constraints, PakTravel can automate daily fleet operations to ensure 100% operational validity while simultaneously optimizing for business preferences like passenger comfort and driver safety.

Part 4: Delay Prediction (Artificial Neural Networks)

Scenario: PakTravel wants to proactively inform passengers, including Ali, if their bus will be delayed. The prediction depends on complex interactions between travel distance, weather, traffic, road quality, and bus age.

Methodology: A synthetic dataset of 500 bus journeys was generated and labelled based on logical rules (e.g., bad weather or heavy traffic automatically causes delays). The data was split (80/20) and normalized using Min-Max scaling. Two Artificial Neural Networks (ANNs) were built:

1. A production-ready model using sklearn's MLPClassifier (two hidden layers: 10 and 8 neurons).
2. A custom model built entirely from scratch using NumPy, implementing forward propagation, binary cross-entropy loss, and backpropagation via gradient descent.

Results: The sklearn ANN achieved outstanding evaluation metrics (Accuracy, Precision, Recall, and F1-score > 90%). The NumPy model successfully trained over 100 epochs, showing a steady decrease in loss. When tested on 5 specific journeys, the model predicted Ali's bus (good weather, good traffic) would be "ON TIME" with high confidence.

Analysis: The high accuracy proves that the hidden layers of the ANN were able to successfully map the non-linear rules dictating bus delays. While the NumPy model performed well, the sklearn model achieved higher accuracy due to its use of the advanced adam optimizer and a higher number of training epochs.

Conclusion: Neural Networks are highly capable of learning complex environmental patterns to predict delays. Implementing this system allows PakTravel to manage customer expectations proactively, leading to higher satisfaction even when logistical disruptions occur.

Part 5: Traveller Clustering (Machine Learning)

Scenario: To increase revenue, PakTravel wants to offer personalized travel packages. However, they first need to discover the natural groupings of their users (Business, Family, and Commuter) based on their travel behaviours.

Methodology: A dataset of 400 user records was generated, containing features like trips per month, booking days, and loyalty points. Exploratory Data Analysis (EDA) confirmed the data was clean. Features were standardized using StandardScaler. The Unsupervised Machine Learning algorithm, **K-Means Clustering**, was applied. The Elbow Method was utilized by testing $k = 1$ through $k = 8$ to determine the optimal number of clusters mathematically.

Results: The K-Means algorithm successfully segregated the 400 travellers into 3 distinct clusters. The Elbow Method produced a graph with a sharp, undeniable bend exactly at $k = 3$, justifying the choice.

Analysis: By analysing the average feature profiles of each cluster, the expected natural groups were clearly identified:

- **Business:** Highest average distance and loyalty points.
- **Family:** Fewest trips per month, medium distances.
- **Commuter:** High frequency of trips, very short distances, and low loyalty points. The scatter plot (Trips vs. Distance) visually confirmed three distinct, well-separated data clouds.

Conclusion: K-Means clustering is a powerful tool for customer segmentation. By mathematically proving the existence of these 3 groups and identifying their core traits, PakTravel's marketing team can now confidently launch highly targeted, personalized promotional campaigns to maximize user retention and revenue.